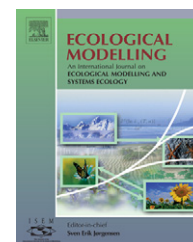


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Modelling forest succession in two southeastern Canadian mixedwood ecosystem types using the ZELIG model

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ABSTRACT

The gap model ZELIG was validated for red spruce–balsam fir–yellow birch and yellow birch–sugar maple–balsam fir forest types in southern Quebec, Canada. Long-term historical data originating from the Lake Edward Experimental Forest, La Mauricie National Park, were used. The effect of the variation in plot size, representing the space within which trees uptake site resources, was also examined. Several species were included in both forest types: red spruce (*Picea rubens* Sarg.), balsam fir (*Abies balsamea* (L.) Mill.), yellow birch (*Betula alleghaniensis* Britton), white birch (*Betula papyrifera* Marsh.), red maple (*Acer rubrum* L.), sugar maple (*Acer saccharum* Marsh.), American beech (*Fagus grandifolia* Ehrh.), eastern hemlock (*Tsuga canadensis* (L.) Carr.) and northern white cedar (*Thuja occidentalis* L.). The pattern of change in basal area growth varied among species, ranging from a steady increase to a more or less rapid decline. There was a good agreement between observations and predictions for yellow birch, red spruce, red maple, sugar maple, balsam fir and northern white cedar. Plot size had a significant impact on the dynamics of the different species. Depending on the species, the decline was accelerated, the amplitude of the fluctuations varied, or the maximum basal area reached changed. Predicted regeneration varied among species and the number of seedlings generally increased with increase in plot size. The pattern of development for most species was related to their life characteristics. The results highlighted the fact that there is a critical lack of knowledge and data on the dynamics of regeneration from the seedling to the sapling stages for the two forest types studied, which resulted in poor predictions for some species. As the life characteristics varied among species, the use of only one plot size for all species may not be realistic.

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1. Introduction

Different types of models have been developed during the last few decades to predict the dynamics of forest ecosystems. The first models were developed in the 1960s and consisted of growth and yield models that predicted tree and stand productivity using variables that could be derived from inventory measurements. Growth and yield models focus on the pre-

diction of the effect of management practices (Mohren and Burkhart, 1994). They are not based on a description of the mechanisms that govern tree and stand growth. The statistical methods used for their calibration aim at estimating the parameters that best fit the model components to observed data. Even though they can predict relatively accurately the short-term effects of silvicultural treatments, they cannot be used safely to predict the growth of forest ecosystems with

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characteristics that extend beyond the range of the data used for their calibration.

As technology development allowed more precise measurements of processes, such as photosynthesis, process-based models were developed to predict forest productivity by modelling the physiological processes that govern the allocation of carbon within trees and stands (e.g. Running and Coughlan, 1988; Running and Gower, 1991; Peng et al., 2002; Paul et al., 2003). Process-based models are considered as very flexible to predict the effect of disturbances, such as insect infestation or global change, because they focus on the description of the mechanisms that affect tree and stand growth. However, their calibration requires extensive ecophysiological and environmental data (Shugart, 1984; Landsberg, 2003). Thus, it may be difficult to use process-based models when environmental site conditions vary considerably or when several species are involved, as ecophysiological data are not usually measured extensively.

Gap models are a compromise between traditional growth and yield models and process-based models. They are considered as a suitable approach by forest ecologists to predict long-term forest dynamics, as they integrate the effects of climatic and site variables and of intra- and inter-specific competition (Bugmann, 2001). Although they integrate ecophysiological factors, they were developed to remain quite general in application. Also, gap models focus on modelling forest succession by integrating several demographic mechanisms, such as tree establishment, growth and mortality. The majority of gap models that have been developed over the last few decades were derived from the JABOWA and FORET models (Bugmann, 2001).

There is a rich body of literature that focuses on the development of gap models. Recent examples can be found in Gullison and Bourque (2001), He et al. (2002) and Song and Woodcock (2003). However, very few studies have dealt with the validation of forest succession models with historical data (e.g. Lindner et al., 1997; Yaussy, 2000; Mäkelä et al., 2000) because long-term records on the same forest ecosystems are extremely rare (Smith and Urban, 1988; Bugmann, 2001). The overall objective of the present study was to validate the gap model ZELIG for red spruce–balsam fir–yellow birch and yellow birch–sugar maple–balsam fir forest types in southern Quebec, Canada, using long-term historical data. The first specific objective was to examine for both forest types the sensitivity of ZELIG to variation in the size of the area within which individual trees compete for site resources. The second specific objective was to contrast the dynamics of the different species in the two forest types.

2. Materials and methods

2.1. Study area

The forest ecosystem types under study were located at the Lake Edward Experimental Forest (LEEF), which is located within La Mauricie National Park (46°45'N, 72°56'W), one of the national parks of the Government of Canada. This territory belongs to the Great Lakes–St. Lawrence forest region (Rowe, 1972). The landscape is quite variable, ranging from moderate

hills to mountains with a maximum elevation of 431 m. Climatic conditions vary considerably throughout the year. During winter, temperature remains mostly below 0 °C and there is snow from November to April. Summers are characterized by warm temperatures, generally above 20 °C. Mean annual temperature is 2.5 °C, growing degree-days are between 2000 and 2600 for a growing season that lasts between 160 and 170 days. Annual precipitation varies between 900 and 1000 mm (Robitaille and Saucier, 1998).

The LEEF, established in 1918 by the Government of Canada, originally measured about 16 km² in area (Hatcher, 1959; Archambault et al., 2003). In 1918, the objective was to study the growth, mortality and regeneration following harvesting, as problems were identified when different silvicultural treatments were undertaken: insufficient softwood regeneration, increase in hardwood proportion, competition from mountain maple (*Acer spicatum* Lamb.) and other undesirable hardwoods. By 1936, 343 sample plots measuring 405 m² in area were established in a grid network, such that the sample plots were 201 m apart from each other. Measurements were conducted in 1936, 1946, 1956, 1967, 1995, 2001, 2003 and 2004. However, approximately half of the original sample plots were measured between 1994 and 2004. The network of sample plots was representative of most forest types that exist in the region. For the present study, two forest types were examined: the red spruce–balsam fir–yellow birch (RSYB) forest type and the yellow birch–sugar maple–balsam fir (SMYB) forest type (Table 1). Data from five uncut sample plots within each forest type were used for the present study. The forest surveys conducted in the different years differed slightly in their procedures. Before 1994, dbh and the number of live trees with dbh larger than 1.3 cm (0.5 in.) were tallied by 2.54 cm (1 in.) dbh classes. Dead trees larger than 9 cm in dbh were also measured. For the 2001 survey, vegetation and soil characteristics were measured by following the procedures outlined by Saucier (1994) for the description of forest ecosystems.

2.2. Description of ZELIG

ZELIG is a gap model based on the same basic principles as the JABOWA and FORET models (Urban, 1990, 2000; Urban et al., 1991). For the present study, ZELIG was selected because (1) it is an individual-tree model that includes realistic representations of tree and stand growth, (2) remains relatively general for application to different forest types, and (3) is adapted for uneven-aged mixed stands (see Urban, 1990). The main processes simulated by ZELIG include photosynthetic active radiation (PAR) interception, tree mortality, treefall gaps and regeneration establishment. The diameter increment for each tree species is computed by integrating the effect of potential growth reduced by environmental constraints. Potential growth consists of the maximum growth rate that a tree species can achieve under optimal conditions in the absence of competition. Environmental constraints include modifiers that account for the effects of changes in PAR conditions, temperature, soil moisture and nutrients. Individual-tree growth, mortality and regeneration are computed annually. Central to ZELIG is the concept of a grid square network that represents the space within which individual trees uptake site resources (Urban et al., 1991, 1999; Coffin and Urban, 1993). This space,

Table 1 – Basic ecological characteristics of the two forest types under study

	Forest type	
	RSYB	SMYB
Main species	Red spruce (<i>Picea rubens</i> Sarg.) Balsam fir (<i>Abies balsamea</i> (L.) Mill.) Yellow birch (<i>Betula alleghaniensis</i> Britton)	Yellow birch Sugar maple Balsam fir
Associated species	White birch (<i>Betula papyrifera</i> Marsh.) Red maple (<i>Acer rubrum</i> L.) Sugar maple (<i>Acer saccharum</i> Marsh.) American beech (<i>Fagus grandifolia</i> Ehrh.) Eastern hemlock (<i>Tsuga canadensis</i> (L.) Carr.) Northern white cedar (<i>Thuja occidentalis</i> L.) Mountain maple (<i>Acer spicatum</i> Lamb.)	White birch Red spruce Mountain maple Red maple American beech Northern white cedar
Deposit	Undifferentiated till	Undifferentiated till
Humus types	Mor, moder	Mor, moder
Humus thickness (cm)	5–8	5–10
Common soil texture	Sandy loam	Fine sandy loam
Mean dbh for the dominant species ^a (cm)		
Balsam fir	20.2	10.6
Red spruce	23.0	22.2
Yellow birch	29.8	25.9
Sugar maple	–	22.8
Mean height (m)		
Balsam fir	19.0	–
Red spruce	22.1	18.3
Yellow birch	18.9	17.1
Sugar maple	–	20.6

^a For trees with dbh greater than 9 cm and measured in 2003 and 2004 in RSYB and SMYB, respectively.

also known as plot size or zone of influence, integrates the ecological characteristics of each individual tree with respect to seedling establishment, growth potential, mortality and capacity to compete for site resources. The evaluation of the sensitivity of ZELIG to variation in the size of the area within which trees use site resources was conducted by systemically varying plot size.

ZELIG considers that tree death is caused by natural mortality or stress adjusted by a stochastic function and the maximum age for each species (Urban, 1990). The modelling of natural mortality assumes that, for each species, the annual death rate remains constant (1%) throughout its natural life-span and that the number of individuals surviving at the maximum age is also constant. Stress mortality occurs for trees that have an annual growth rate that is less than 10% of their potential growth rate or less than 1 mm for at least two continuous years. For these trees, ZELIG assumes that only 1% of them will survive 10 years. Thus, for both types of mortality, the original version of ZELIG assumes that the basic mortality rate is the same for all species. The first simulations in the present study indicated that this assumption for natural mortality was not realistic for the forest types under study. Therefore, ZELIG was modified accordingly to allow each species to have its own basic death rate associated with natural mortality. Species-specific death rates were estimated by analyzing permanent sample plots of the Forest Inventory Service of the Quebec Ministry of Natural Resources and Wildlife (ministère des Ressources naturelles et de la Faune du Québec). Stress and mortality factors used for each species are listed in Table 2.

Regeneration is modelled as a stochastic event and is influenced by site conditions. First, a site-specific relative seedling establishment rate (RSER) is provided to ZELIG for each species. This variable, which represents the number of seedlings that will likely germinate and reach breast height under optimal conditions, is used to rank the species with respect to their regeneration potential. Then, RSER is adjusted to account for the effects of available light conditions, soil moisture and fertility, species-specific fertility response class and degree-days. For the present study, RSER for each species was estimated by undertaking regeneration surveys in both forest types in 2003 and 2004 (Table 2).

ZELIG requires the following information for each species in the forests under simulation: maximum age, dbh and height, minimum and maximum growing degree-days, and tolerance class for shade, nutrient stress and drought (Urban, 1990). For the species present in the study area, the information is included in Table 2 and was obtained from the files included in one of the ZELIG web sites (<http://www.eco.wiz.uni-kassel.de/model.db/mdb/zelig.html>) and from Botkin et al. (1972). ZELIG allows the user to initiate the simulations from bareground or to provide individual-tree data. For the present study, individual-tree data based on the first measurement within each plot, 1936 or 1946, were used to initiate ZELIG (Table 3). Predictions obtained for each sample plot at different time cycles were matched with individual-plot data on the basis of annual time difference since first measurement. Thus, observations in 1936 or 1946, 1956, 1967 and 2003 or 2004 were matched with the corresponding time cycle of the predictions. For conciseness, plot data were aver-

Table 2 – Species-specific allometric and ecological characteristics provided to ZELIG for the two forest types at the Mauricie National Park (adapted from http://www.eco.wiz.uni-kassel.de/model_db/mdb/zelig.html and from Botkin et al. (1972))

Species	Amax	Dmax	Hmax	g	Dgmin	Dgmax	Ligtol	Mdrg	Fert	RSER	StrMort	NatMort
Yellow birch	250	122.0	3050	100	1420	3084	3	2	2	0.25	0.369	2.500
White birch	140	100.0	2500	160	700	2500	4	3	3	0.04	0.369	1.000
Red spruce	300	100.0	3000	89	500	2580	2	2	3	0.275	0.369	1.000
Mountain maple	50	18.0	800	150	889	5500	1	2	3	11.5	0.369	4.733
Striped maple	75	22.5	1000	150	889	5500	1	2	3	0.05	0.369	4.439
Red maple	150	100.0	3000	176	1260	6601	2	3	3	1.5	0.369	0.500
Sugar maple	300	150.0	4011	89	1204	3200	1	2	2	8.75	0.369	1.000
American beech	366	122.0	3660	72	1327	5556	1	2	2	0.01	0.369	4.919
Eastern hemlock	650	150.0	3500	47	1324	3100	1	2	3	0.01	0.369	4.678
Balsam fir	200	50.0	1500	69	250	2404	1	1	3	2.25	0.369	0.500
Northern white cedar	400	100.0	2400	55	1000	2188	2	4	3	12.5	0.369	0.500
Hobblebush	50	18	200	158	1928	6011	5	3	2	0.25	0.369	4.961

Note: Amax, maximum age for the species; Dmax, maximum diameter at breast height (cm); Hmax, maximum height (m); g, growth rate scaling coefficient; Dgmin, minimum growing degree-days; Dgmax, maximum growing degree-days; Ligtol, shade tolerance class; Mdrg, maximum drought tolerance class; Fert, soil fertility response class; RSER, relative seedling establishment rate. (RSER may vary with site-specific conditions) StrMort, probability factor of stress mortality; NatMort, probability factor of natural mortality.

Table 3 – Mean dbh and height to live crown for the different species present at the first measurement in 1936 or 1946 in the sample plot data provided to ZELIG to simulate the succession in red spruce–balsam fir–yellow birch (RSYB) and yellow birch–sugar maple–balsam fir (SMYB) forest types in the La Mauricie National Park

Species	RSYB		SMYB	
	dbh (cm)	Height to live crown (m)	dbh (cm)	Height to live crown (m)
Yellow birch	15.7	4.6	20.4	5.1
White birch	11.9	6.0	12.8	5.6
Red spruce	8.4	4.0	7.6	4.0
Red maple	13.1	5.6	10.0	4.6
Balsam fir	10.2	3.9	11.6	4.1
Northern white cedar	8.4	1.9	6.7	2.0
Eastern hemlock	25.4	3.0		
Sugar maple	9.7	2.5	9.7	2.5
American beech			19.9	3.5

aged for each forest type studied. Predicted basal area was compared with observed basal area by computing the mean percent of absolute annual difference between observations and predictions (Vanclay, 1994):

$$\text{MPOP} = \frac{\sum (\hat{y}_i - y_i / |y_i|)}{n} \times 100 \quad (1)$$

where y_i is observed basal area and $\hat{y}_i$ the predicted basal area.

3. Results

There were three patterns of change in predicted basal area over time in RSYB (Figs. 1–3). Yellow birch and balsam fir declined more or less rapidly and disappeared or had very low basal area thereafter. Basal area for white birch and eastern hemlock increased for a certain period, and then declined more or less rapidly. Red spruce is the only species that increased steadily over time in each plot size. Red maple increased steadily over time, but only in the 200 m² plot size. For the two other plot sizes, the pattern of change for red maple was characterized by an increase early in the simulation, followed by a more or less rapid decline. Basal area for

northern white cedar fluctuated, but did not change appreciably. The variation in plot size had an impact on the pattern of response for most species in RSYB. The increase in plot size changed the occurrence of the disappearance of yellow birch and eastern hemlock, delayed the disappearance of white birch, reduced maximum basal area obtained for balsam fir, red spruce and northern white cedar, and modified the pattern of change for red maple. Plot size also had an effect on MPOP for all species (Table 4). There was a decrease in MPOP with increase in plot size for yellow birch, red spruce, red maple, balsam fir and northern white cedar. For white birch and eastern hemlock, MPOP increased from the 200 to the 300 m² plot size, and then decreased from the 300 to the 400 m² plot size.

As the effect of variation in plot size was important and varied among the species in RSYB, an optimum plot size was determined on the basis of the maintenance of the species in term of basal area and/or differences between observations and predictions. The 400 m² plot size was optimum for yellow birch, red maple, balsam fir and northern white cedar. For yellow birch and balsam fir, the lowest differences between observed and predicted basal areas were obtained for nearly all the years under this plot size. In particular, simulated basal area for yellow birch in the 400 m² plot

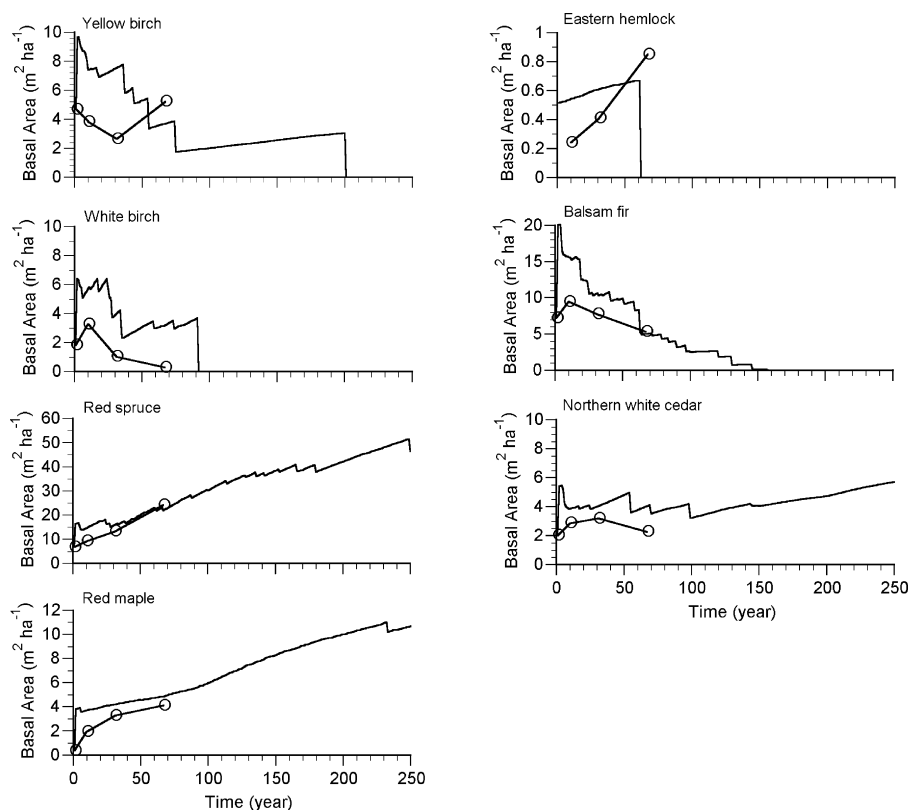


Fig. 1 – Mean predicted and observed basal area ($\text{m}^2 \text{ha}^{-1}$) for species in the red spruce–balsam fir–yellow birch forest type using a 200 m^2 plot size (The circle symbols represent observations).

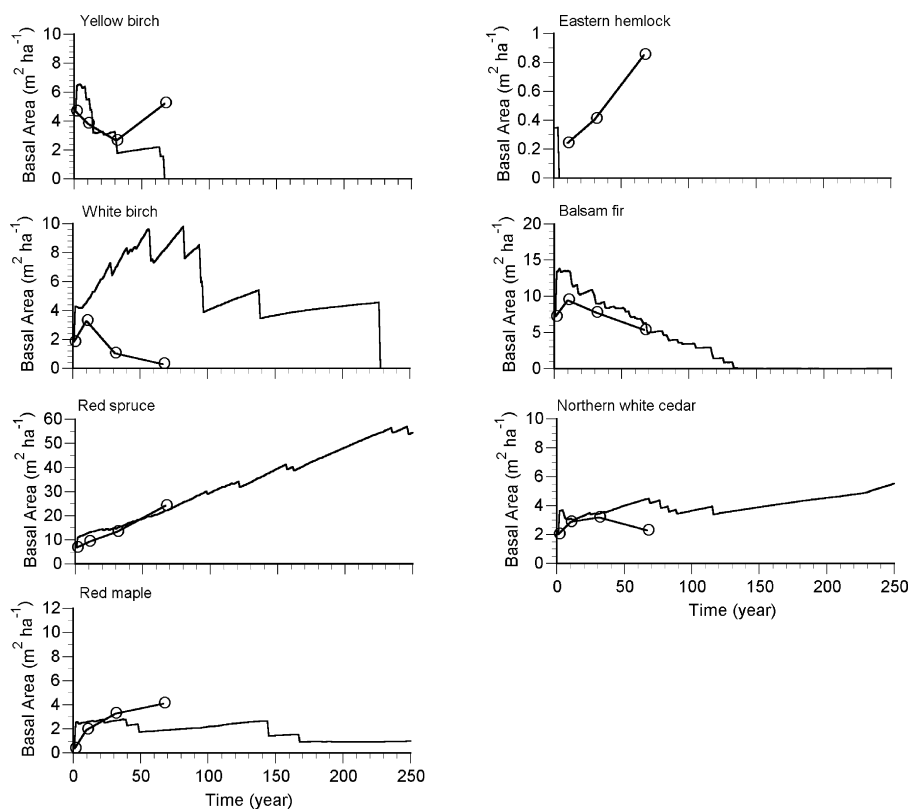


Fig. 2 – Mean predicted and observed basal area ($\text{m}^2 \text{ha}^{-1}$) for species in the red spruce–balsam fir–yellow birch forest type using a 300 m^2 plot size (The circle symbols represent observations).

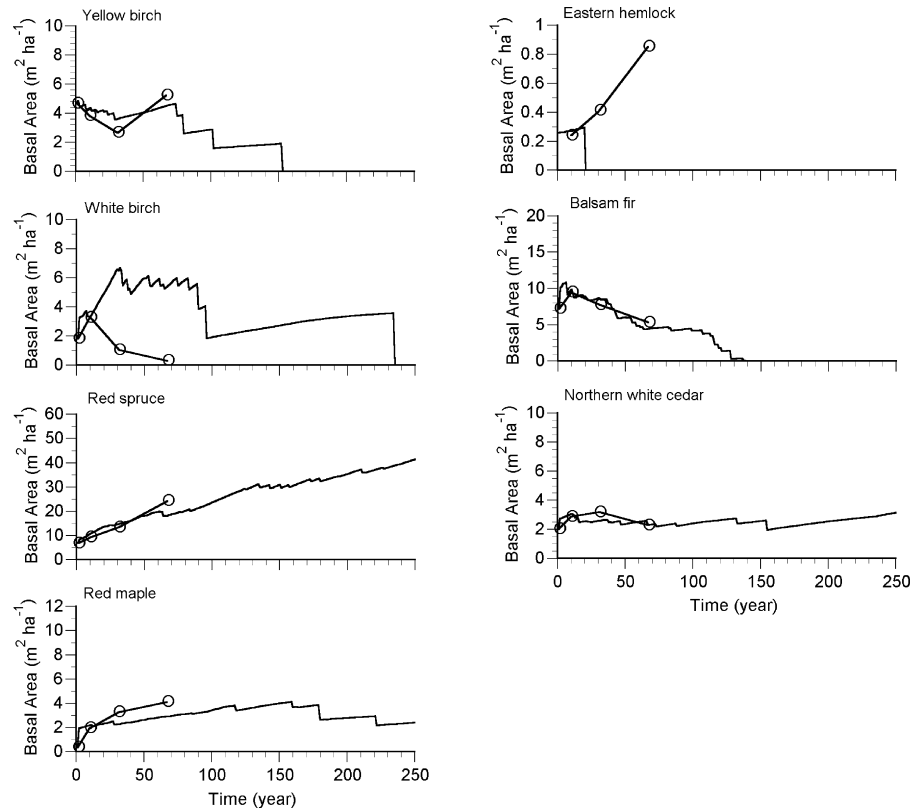


Fig. 3 – Mean predicted and observed basal area ($\text{m}^2 \text{ha}^{-1}$) for species in the red spruce–balsam fir–yellow birch forest type using 400 m^2 plot size (The circle symbols represent observations.).

size represented the closest pattern of change obtained for observed basal area: small decrease in basal area in the first 40 years, followed by an increase. However, yellow birch disappeared earlier in the 400 m^2 plot size than in the 200 m^2 plot size. Simulated basal area for red maple in the 400 m^2 plot size had the closest match with the long-term pattern of change in observed basal area. For northern white cedar, there was very close agreement between observed and predicted basal areas and pattern of change: relatively small increase in basal area early in the simulation followed by a small decline. The 200 m^2 plot size was optimum for white birch. Despite pronounced differences between observed and pre-

dicted basal areas, the simulated basal area in the 200 m^2 plot size most closely matched the long-term pattern of change in observed basal area. For red spruce, all three plot sizes could be considered as optimum, as there was very close agreement between observed and predicted basal areas in all of them. Also, red spruce increased steadily over time in the three plot sizes. The pattern of change in simulated basal area that approximated the closest the observed pattern for eastern hemlock took place in the 200 m^2 spacing. Even though the 200 m^2 spacing did not have the lowest MPOP, it is in this plot size that the pattern of change in simulated basal area matched the closest the pattern for observed basal area

Table 4 – Mean percent of absolute annual difference (MPOP) between observed and predicted basal area obtained for the different plot sizes in the red spruce–balsam fir–yellow birch (RSYB) and the yellow birch–sugar maple–balsam fir (SMYB) forest types

	RSYB, plot size (m^2)			SMYB, plot size (m^2)		
	200	300	400	200	300	400
Yellow birch	102	48	21	60	10	24
White birch	464	1130	788	735	604	356
Red spruce	29	20	17	226	142	111
Red maple	42	35	24	46	17	8
Eastern hemlock	91	100	70			
Sugar maple				7	13	21
American beech				77	58	58
Balsam fir	36	26	10	406	576	877
Northern white cedar	46	38	9	77	67	94

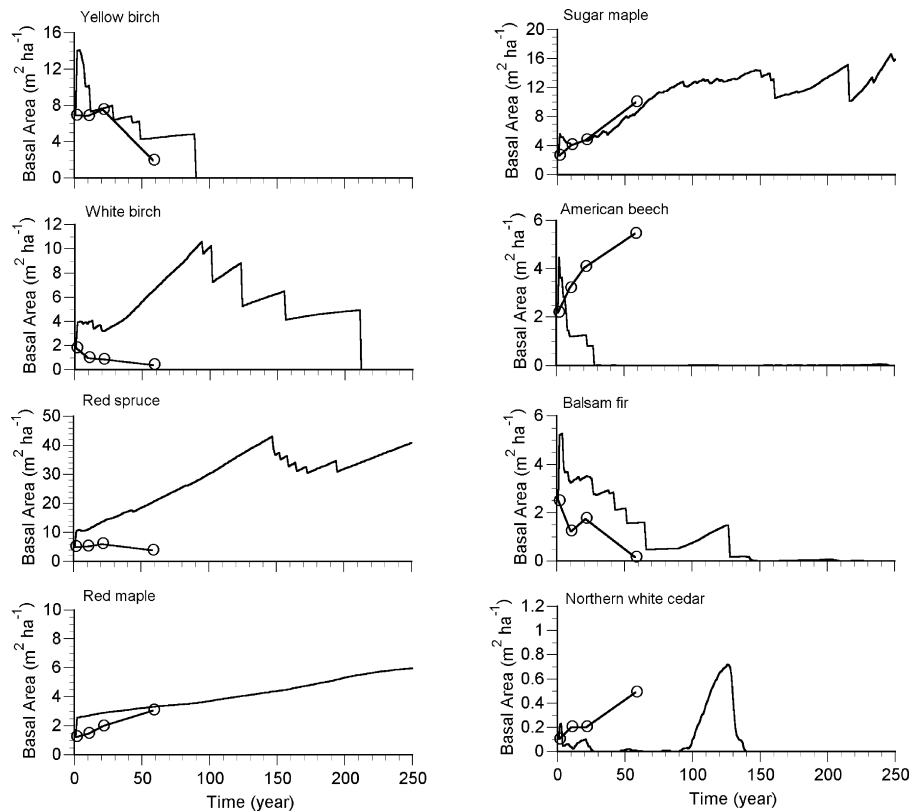


Fig. 4 – Mean predicted and observed basal area ($\text{m}^2 \text{ha}^{-1}$) for species in the yellow birch–sugar maple–balsam fir forest type using a 200 m^2 plot size (The circle symbols represent observations.).

and that eastern hemlock remained present for the longest period.

Compared with RSYB, there were similar patterns of change in predicted basal area in SMYB (Figs. 4–6). Yellow birch, American beech and balsam fir declined more or less rapidly and disappeared or reached very low basal area. A decline was also obtained for white birch, but after a period of increase in basal area. Despite fluctuations, red spruce and sugar maple had a pattern of increase in basal area. Northern white cedar was characterized by a cyclic pattern. Red maple was the only species that increased in basal area. Plot size had an effect on the pattern of change in basal area for most species. As plot size increased, the declines of yellow birch, American beech and balsam fir were delayed and the amplitude of the fluctuations in predicted basal area of white birch decreased from the 200 to the 300 m^2 plot size and increased from the 300 to the 400 m^2 plot size. The maximum basal area reached by red spruce decreased from the 200 to the 300 m^2 plot size. For red maple, the only noticeable effect of increase in plot size consisted in the increase in predicted maximum basal area from the 300 to the 400 m^2 plot size. Sugar maple had a pattern of increase similar to that of red spruce. However, the maximum basal area reached remained fairly close in the 200 and 300 m^2 plot sizes, and decreased from the 300 to the 400 m^2 plot size. For northern white cedar, the amplitude of the cycles increased from the 200 to the 300 m^2 plot size and decreased from the 300 to the 400 m^2 plot size. MPOP also changed with plot size for all species (Table 4). As plot size

was increased, MPOP decreased for white birch, red spruce and red maple. A similar pattern was obtained for American beech, except that the decrease took place only from the 200 to the 300 m^2 plot size. Yellow birch and northern white cedar had the lowest MPOP in the 300 m^2 plot size. For sugar maple and balsam fir, MPOP increased with increase in plot size.

Not all the species had the same optimum plot size in SMYB. The 200 m^2 plot size was optimal for sugar maple and balsam fir. For sugar maple, this plot size resulted in the most consistent long-term pattern of predicted versus observed basal area and the lowest MPOP (Table 4). Even though there were large differences in MPOP for balsam fir, there was relatively good agreement between the long-term patterns in predicted and observed basal areas in the 200 m^2 plot size relative to the other plot sizes. Based on MPOP only, the 300 m^2 plot size could be considered as the optimum plot size for yellow birch and northern white cedar. For yellow birch, the 300 m^2 plot size was optimum both in terms of consistency in the long-term pattern and differences between predictions and observations. Even though the lowest MPOP for northern white cedar was obtained in the 300 m^2 plot size, the long-term patterns outlined by the predicted and observed basal areas differed considerably. For instance, observations indicated an increase in basal area for northern white cedar, while predicted basal area increased for the first 30 years and decreased thereafter. The optimum plot size for this species is probably between 300 and 400 m^2 . The 400 m^2 plot size was optimum for red maple both in terms of minimal MPOP and agreement

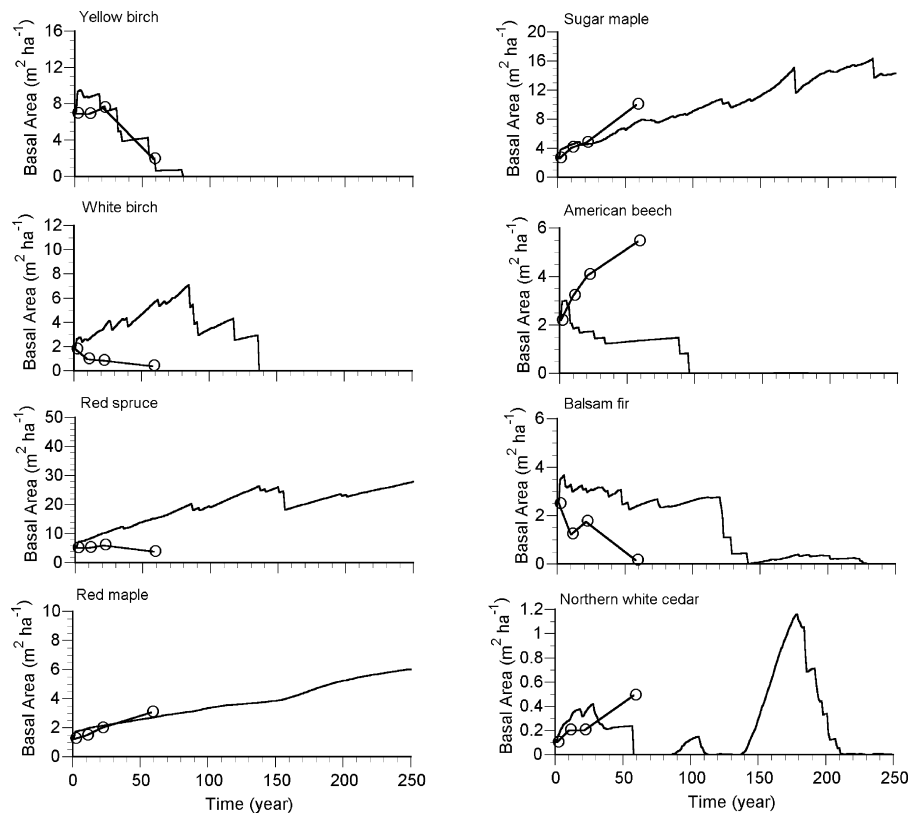


Fig. 5 – Mean predicted and observed basal area ($\text{m}^2 \text{ha}^{-1}$) for species in the yellow birch–sugar maple–balsam fir forest type using a 300 m^2 plot size (The circle symbols represent observations.).

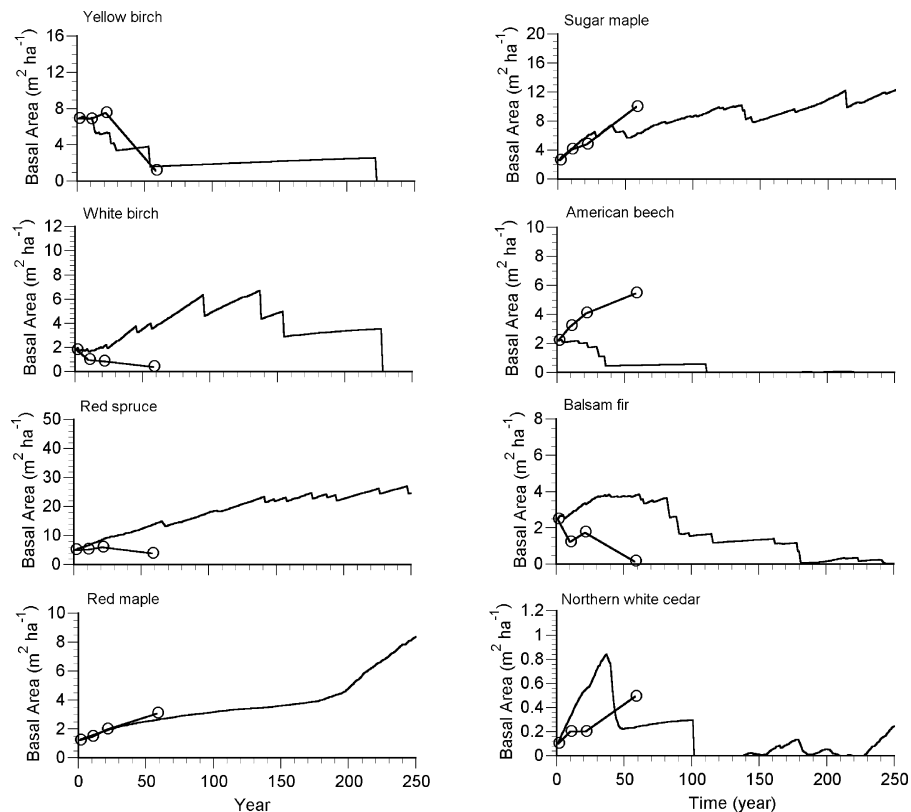


Fig. 6 – Mean predicted and observed basal area ($\text{m}^2 \text{ha}^{-1}$) for species in the yellow birch–sugar maple–balsam fir forest type using a 400 m^2 plot size (The circle symbols represent observations.).

Table 5 – Predicted regeneration (number of seedlings m^{-2}) at the end of a 250-year simulation period in the different plot sizes in the red spruce–balsam fir–yellow birch forest type

	Plot size (m^2)		
	200	300	400
Yellow birch	0.8	0.4	0.6
White birch	0.4	0.6	2.6
Red spruce	42.6	56.6	80.4
Red maple	12.4	56.0	80.2
Balsam fir	72.6	66.2	258.4
Northern white cedar	2.0	10.4	14.6

Table 6 – Predicted regeneration (number of seedlings m^{-2}) at the end of a 250-year simulation period in the different plot sizes in the yellow birch–sugar maple–balsam fir forest type

	Plot size (m^2)		
	200	300	400
Yellow birch	0.2	0.4	0.8
White birch	0.2	0	0
Red spruce	0.8	0.6	2.6
Red maple	10.4	11.4	24.8
Sugar maple	151.8	158.2	379.2
American beech	5.4	10.2	19.2
Balsam fir	12.0	9.6	20.2
Northern white cedar	39.6	23.8	32.4

between long-term changes in observed and predicted basal areas. For white birch, red spruce and American beech, the 400 m^2 plot size could be considered as optimum on the basis of low MPOP, particularly for red spruce. However, there was no optimal plot size for the three species because the predicted and observed long-term patterns of change diverged completely. For instance, predicted basal area for white birch and red spruce increased for a long period, but observations indicated a decrease. Predicted basal area for American beech decreased, but observations indicated an increase.

The number of seedlings generated at the end of the 250-year simulations differed among the species in RSYB (Table 5). Red spruce, red maple and balsam fir generated many seedlings compared with the other species. Plot size had an important effect on regeneration for some species. While the change in plot size resulted in minor variations in the number of seedlings generated for yellow birch, the number of seedlings generated increased significantly from the 200 to the 400 m^2 plot size for white birch, red spruce, red maple, balsam fir and northern white cedar.

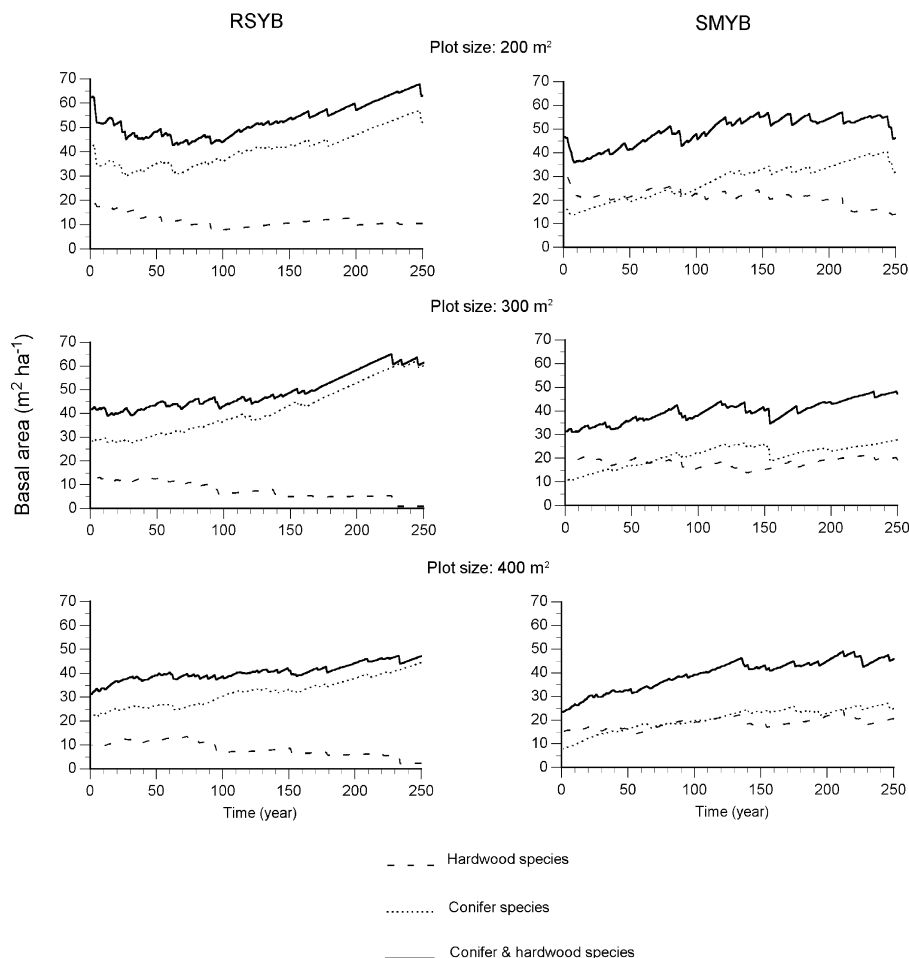


Fig. 7 – Mean predicted basal area for conifer and hardwood species for plot sizes varying from 200 to 400 m^2 in the red spruce–balsam fir–yellow birch (RSYB) and yellow birch–sugar maple–balsam fir (SMYB) forest types.

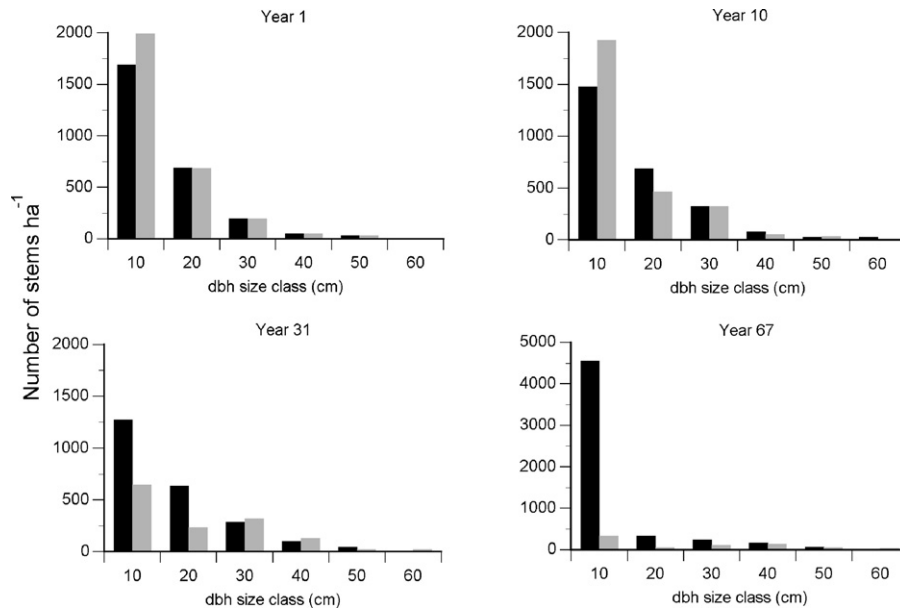


Fig. 8 – Comparison of observed (black) and predicted (gray) dbh distributions in the red spruce–balsam fir–yellow birch (RSYB) forest type at different projection years.

The patterns of predicted regeneration in SMYB differed from the pattern in RSYB for some species (Table 6). Northern white cedar generated many more seedlings at the end of the 250-year simulation period in comparison with those generated in RSYB. On the other hand, many times fewer seedlings were generated in SMYB for red spruce, red maple and balsam fir than in RSYB. Yellow birch and white birch had very few seedlings generated and their number did not change much with plot size. The greatest number of seedlings predicted was for sugar maple. For all the species, there was a pattern of

increase in the number of seedlings generated with increase in plot size, except for white birch, balsam fir and northern white cedar.

The comparison of predicted total basal area for hardwood and conifer species indicated differences among both forest types (Fig. 7). For RSYB, there was a pattern of decrease in basal area for both hardwood and conifer species with increase in plot size, particularly from the 300 to the 400 m² plot size. Independently of plot size, the proportion of hardwood species decreased, while the proportion of conifer species increased.

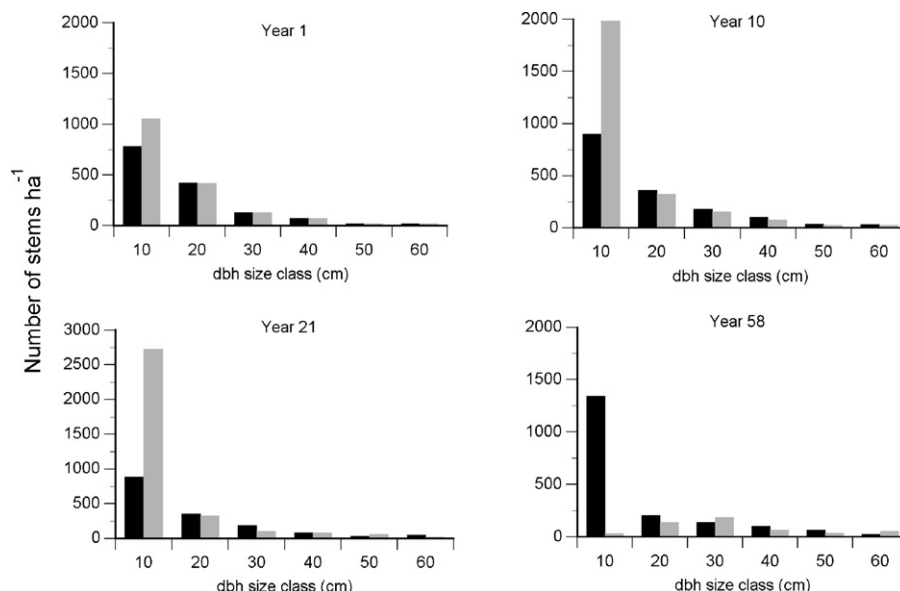


Fig. 9 – Comparison of observed (black) and predicted (gray) dbh distributions in the yellow birch–sugar maple–balsam fir forest type (SMYB) at different projection years.

For SMYB, predicted basal area for conifer species decreased with increase in plot size, while basal area for hardwood species did not change appreciably.

For conciseness, simulated dbh distributions were compared with observed dbh distributions by using the results of the simulations for the 400 m² plot size for both forest types. The dbh size distribution in RSYB was generally well predicted in years 1 and 10, except for the 10 cm dbh size class where differences between observed and predicted densities varied from 300 to 500 stems per ha (Fig. 8). For years 31 and 67, there were major differences between observations and predictions in the two smallest dbh classes. For SMYB, there was a relatively good agreement between observed and predicted numbers of trees, but only for the dbh classes greater than or equal to 20 cm (Fig. 9). There were major discrepancies for all years in the 10 cm dbh class.

4. Discussion

The effects of plot size on the predicted dynamics of the stands were consistent with those observed by Coffin and Urban (1993) with respect to the pattern of decrease in predicted stand basal area with increase in plot size. They attributed this pattern to a dilution effect. As plot size increases, large trees occupy a relatively small area within the plot such that the “resource space” is not much affected, even if mortality occurs. The results of the present study also indicated that plot size had a significant effect on predicted regeneration. For nearly all species, the number of seedlings predicted increased with increase in plot size. This pattern can be attributed to the fact that ZELIG simulated more constraints for the establishment and survival of seedlings in small plot sizes (Coffin and Urban, 1993). The differences in the life-history characteristics of the species also explain why they did not respond similarly to changes in plot size.

Red spruce was the most stable species in RSYB, being characterized by a regular pattern of increase in both predicted and observed basal area in the three plot sizes (Figs. 1–3). For SMYB, a similar pattern was observed for red spruce, but also for red maple and sugar maple (Figs. 4–6). Change in plot size only affected the maximum basal area reached or induced fluctuations. The stable behavior observed was consistent with their biology. The three species are characterized by good seedling establishment rate and are classified as shade tolerant (Blum, 1990; Walters and Yawney, 1990; Godman et al., 1990). As it is a long-lived species, red spruce contributes to the maintenance of long-term stability in association with species such as balsam fir (Ray, 1956; Busing and Wu, 1990). However, on these sites, red spruce seedling density may be low in comparison with other species such as balsam fir (Seymour, 1992; Brissette, 1996).

Compared with red spruce, red maple and sugar maple, predicted basal area for the other species changed more or less regularly over time. Yellow birch fluctuated and declined in both forest types and was quite sensitive to change in plot size (Figs. 1–6). This pattern may appear consistent because yellow birch is a mid-tolerant species and its crown requires space to expand so that it can grow adequately (Erdmann, 1990). The fact that ZELIG predicted its disappearance in a rel-

atively short period of time relative to other species can be explained by its poor competitive ability and poor regeneration capacity. However, it may also be argued that its disappearance is not realistic because yellow birch has been identified for a long time as a climax species in these ecosystems (e.g. Heimbürger, 1941; Gosselin, 2001). The fact that ZELIG predicted poor regeneration for yellow birch is consistent with its life history. In particular, yellow birch is known for the poor establishment rate of its seedlings (e.g. Bellefleur and Larocque, 1983; Houle and Payette, 1990; Roberts and Dong, 1993; Hewitt and Kellman, 2004).

Predicted basal area for white birch increased early in the simulations in both forest types. This increase was followed by a pattern of decline in all plot sizes for both forest types. This was consistent with its life history. White birch is a short-lived intolerant transitional species (Safford et al., 1990). ZELIG always predicted poor regeneration for both forest types (Tables 4 and 5). These results were also consistent with the life history of white birch: despite the fact that seed production may be high, the proportion of viable seeds may be low and young seedlings are very sensitive to environmental stress (Safford et al., 1990). Balsam fir in both forest types and American beech in SMYB were also characterized by a pattern of decline in predicted basal area, and variation in plot size simply affected the rate of decline or induced a cyclic pattern (Figs. 1–6). For balsam fir, this predicted pattern was consistent with observations. However, the fact that it disappeared before the end of the 250-year simulation period is unrealistic, as balsam fir is a climax species for both forest types. Its predicted disappearance might be related to the poor prediction of seedling survival. Balsam fir was among the species for which ZELIG predicted a high number of seedlings (Tables 4 and 5). These results suggest that there was a problem in the prediction of the transition from the seedling stage to the tree stage by ZELIG. The rapid decline of American beech in SMYB was not consistent with observed data. Also, this behavior was not consistent with its life history. American beech is considered a shade tolerant species with a good seedling establishment rate (Tubbs and Houston, 1990).

The differences between predicted and observed dbh distributions were important mostly in the lowest dbh class, which contained trees that were less than 10 cm in dbh, but greater than 2.5 cm. These results suggest that ZELIG did not predict well the transition from the seedling to the sapling stage (when trees reach a dbh of 2.5 cm). This is a crucial point, as it was estimated that sapling density in both forest types was adequate to maintain the long-term stability in terms of species composition (see Archambault et al., 1998, 2003). In addition, this situation may contribute to diminishing the capacity of ZELIG to predict realistically the long-term succession in both forest types, particularly if one considers that ZELIG has been used to simulate forest dynamics for periods as long as 500 or 1000 years (e.g. Coffin and Urban, 1993; Miller and Urban, 1999). Therefore, there is a need to integrate in ZELIG a better understanding of the dynamics of regeneration in terms of growth and mortality.

Changes in plot size had a significant effect on the dynamics of most species and the proportion of conifers and hardwoods for the simulation period considered in the present study. The results also indicated that the optimum plot size

varied among the species. However, the optimum plot size estimated above for each species did not necessarily coincide with the plot size that was most favorable to seedling establishment, which suggests that the use of a single plot size for all the species did not represent sufficiently well the biological reality. Each species has its own ecological characteristics with respect to resource requirement. Their crown characteristics differ in terms of space occupancy and light interception capacity. As the extension and uptake rate of their root systems differ, they create differential demands on soil resources. Also, the amplitude of the zone of influence increases over time as trees become bigger (Weishampel et al., 1992). Regarding species-specific effects on seedling establishment, there are notable differences in seed production and viability and propagation rate and intensity. The use of the same plot size for all the species in the present study resulted in an additional problem. For the 200 and 300 m² plot sizes, predicted stand basal areas were unrealistically high for several ages in both forest types. On the other hand, these two plot sizes were also considered as optimal plot sizes for some species.

A better representation of the differences in ecological characteristics among the species may be an incentive to integrate more physiological traits in a gap model such as ZELIG. However, this must be done with caution with gap models. As several species are generally included in this type of model, it is not evident that the physiological characteristics are sufficiently well known for all the species such that they can be easily used in gap models (Bugmann, 2001). Thus, a refinement of the life characteristics or functional types, such as those used in Table 2, of the different species involved must be considered to ensure that their behaviors with respect to the use of site resources are better differentiated. This can contribute to increasing the degree of realism and precision of gap models. For the tree layer, this refinement may consist in increasing the number of tolerance classes and further differentiating the functional types of the environmental response functions. For the seedling layer, improved representations of the biological factors that influence seed production and propagation and of the effects of site conditions on their development are desirable.

5. Conclusion

Gap models have been considered mostly as academic tools used by ecologists to better understand the dynamics of forest ecosystems. However, they are becoming more and more necessary to ensure the sustainability of mixedwood uneven-aged forest ecosystems. The traditional growth and yield models used by forest managers have focused on the prediction of the growth of the dominant tree layers in forest stands for relatively short periods of time, usually until harvesting. Most of these models do not have the capacity to predict seedling establishment and growth and the dynamics following clearcutting or the application of different types of management treatments. As the planning of the maintenance of sustainable forest management requires a better understanding of the dynamics of forest stands from the seedling stage to maturity and beyond, gap models are conceptually appropriate to meet the prediction needs of forest managers.

However, compared with traditional growth models, their validation is more problematic, as it is not always possible to compare their predictions with reality using historical data. The present study, which used historical data, indicated that ZELIG predicted relatively well the dynamics of several species in two mixedwood forest ecosystem types in southern Quebec. However, a better integration of life characteristics of different species and a better understanding of the dynamics of the transition from the seedling to the tree stage are required to improve its predictive capacity for the mixedwood forest types that were used in the present study. In particular, the concept of plot size must be adaptable such that it better represents the differences among species with respect to the use of resources.

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